

LESSON 7.5

ADDING INTEGERS



LESSON INTRODUCTION

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

LEARNING TARGETS

- I can add integers with procedural fluency.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What are additive inverses?
- How can you use the signs of addends and their absolute values to determine the sign of their sum?
- Describe how to add integers with the same signs and with different signs?

LESSON NARRATIVE

In this lesson, students begin by modeling additive inverses on number lines and with integer chips. Students discover why the sign of the sum stays the same when the signs of the addends are the same. They also discover why, when the signs of the addends are not the same, the answer is the amount remaining after canceling the zero pairs with the sign of the addend with the greater absolute value.

COMPONENT SUMMARY

7.5.1 WARM-UP (~5 MIN.)

Multiple representations

7.5.3 GUIDED INSTRUCTION (10 MIN.)

Connect additive inverses to zero pairs for adding with the same or opposite signs

7.5.5 COLLABORATION (10 MIN.)

Fill in missing addends and sums in integer addition equations

7.5.7 WRAP-UP (~5 MIN.)

Reflect on what they are still unsure about when adding integers

7.5.2 EXPLORATION (10 MIN.)

Explore the concept of additive inverses

7.5.4 TRY IT (5 MIN.)

Practice adding integers with the same and different signs

7.5.6 YOUR TURN! (5 MIN.)

Practice adding integers with the same and different signs

RESOURCES

GLOSSARY

Key Terms:

- Existence of additive inverses

Additional Terms:

- Absolute value
- Integer

MATERIALS

- Integer chips

ADDITIONAL PREPARATION

- If differentiating problems for 7.5.4 Try It or 7.5.6 Your Turn!, prepare problems to replace and supplement the problems in the lesson.
- Consider creating an anchor chart to illustrate how to add integers with the same and with different signs. Include instructions, visual models (number line and integer chips), and a few examples.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to explain how many zero pairs are created in 7.5.3 Guided Instruction.
- Students may have difficulty finding the missing addends in 7.5.5 Collaboration.

THINGS TO CONSIDER

- Consider displaying an anchor chart that illustrates how to add integers with the same and different signs. Include instructions, visual models (number line and integer chips), and a few examples.
- Consider rewriting expressions in 7.5.3 Guided Instruction to show the additive inverses involved as students develop procedural fluency when adding without manipulatives or number lines.
- Consider allowing students to use the models that make sense to them to add integers if they are not yet at a place to do so without. Encourage the use of number-line models in this case because of the size of the addends in the lesson.

LITERACY AND LANGUAGE ROUTINES

Take Turns

In 7.5.5 Collaboration, consider having students take turns explaining how they added the integers with different signs. These discussions will improve student understanding and confidence.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

7.5.5 Collaboration provides an opportunity for students to engage in mathematical discussion through explaining their mathematical process and justifying their results to a peer. Use this as an opportunity for students to practice articulating their mathematical thinking and comparing their methods to others.

DIFFERENTIATE INSTRUCTION

Each activity in this lesson provides opportunity to differentiate instruction by content, by process, or by product.

- Provide integer chips for 7.5.2 Exploration and 7.5.5 Collaboration for kinesthetic and visual entry points.
- Offer verbal responses for students to submit via phones or apps for the 7.5.7 Wrap-Up.
- Consider allowing students to “CYOI,” or “Choose Your Own Integer,” for 7.5.4 Try It and 7.5.6 Your Turn!. Students who need lower entry points can choose numbers they feel confident with exploring, while students who are ready for a challenge can choose larger numbers.

7.5.1 WARM-UP: REPRESENTING ADDITION

Component Overview: Students write and evaluate addition expressions represented by integer chips and a number-line model.

ENRICHMENT

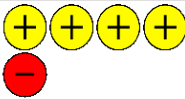
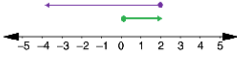
Based on your observations from Lesson 4, consider using prompting questions for students who need additional support.

- How did you determine how to write the addition expression represented by the integer chip model?
- How can you use integer chips to help you find the sum?
- How can you represent the expression using a number line?
- What does each ray on the number line represent?
- How did you determine how to write the addition expression represented by the number-line model?
- Why is the sum negative?
- How can you represent the expression using integer chips?



7.5.1 Warm-Up: Representing Addition

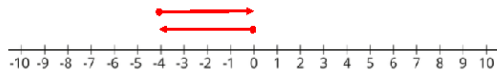
1. Write the addition expression represented by the model. Then, find the sum.

Model	Addition Expression	Sum
	$4 + (-1)$	3
	$2 + (-6)$	-4



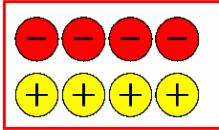
7.5.2 Exploration: Additive Inverses

1. Use the number line to show the sum of $-4 + 4$. Then, complete the equation.

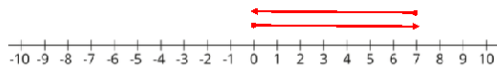


$$-4 + 4 = 0$$

2. Represent the sum of $-4 + 4$ using integer chips.

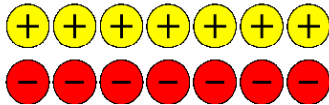


3. Use the number line to show the sum of $7 + (-7)$. Then, complete the equation.



$$7 + (-7) = 0$$

4. Represent the sum of $7 + (-7)$ using integer chips.



Existence of Additive Inverses

For every a there exists $(-a)$ so that $a + (-a) = 0$ and $(-a) + a = 0$.

7.5.2 EXPLORATION: ADDITIVE INVERSES

Component Overview: Students explore multiple representations of additive inverses and finding their sum. Students see additive inverses represented on number lines and with integer chips. Students connect additive inverses to zero pairs. Finally, students create an example of an additive inverse and share their example with a partner.

ENRICHMENT

Consider using question stems for students to answer on their own or for you to ask while circulating.

- What do you notice about the sum of each question?
- What do you notice about the values being added?
- Do additive inverses remind you of anything you learned about when modeling integers with integer chips?

Additive inverses are another way to represent the concept of a zero pair.

5. What number is the additive inverse of -15 ?

15

6. What number is the additive inverse of 27 ?

-27

7. Choose any positive integer and write it in the blank:
Answers will vary. Any positive integer is correct. The example provided here uses 12.

Use your number to fill in the blanks in the statement.

For the value 12, there exists (-12) so that

$$12 + (-12) = 0 \text{ and } (-12) + 12 = 0.$$

8. Write your partner's positive integer:

Answers will vary. The example provided here uses 9.

Use your partner's number to fill in the blanks in the statement.

For the value 9, there exists (-9) so that

$$9 + (-9) = 0 \text{ and } (-9) + 9 = 0.$$

ENRICHMENT

How do you know that is the additive inverse of ?

7.5.3 GUIDED INSTRUCTION: ADDING INTEGERS

Component Overview: Students summarize concepts explored in previous lessons adding integers and relate these concepts to how to think about adding integers without using manipulatives or number-line models. They formalize this learning as they continue developing procedural fluency.

ENRICHMENT

How does the number of zero pairs formed relate to additive inverses?

TEACHER MOVES

Consider rewriting the expression as $40 + (-40) + (-125)$ and think aloud to compare the two expressions and discuss how this shows how the additive inverse of 40 can be used to consider the zero pairs created and eliminated when adding $40 + (-165)$. The chips that would remain after eliminating the zero pairs are 125 negative chips, or -125.

DIFFERENTIATE INSTRUCTION

If students struggle to understand why 40 zero pairs are created, consider modeling a problem with smaller values to model how the zero pairs are created using integer chips.

TEACHER MOVES

Utilize talk moves to position students for synthesis.

“Karma, in your own words, how would you describe the way to use zero pairs for adding integers? Chantoy, restate Karma’s approach but in your own words.”



7.5.3 Guided Instruction: Adding Integers

Understanding the existence of additive inverses can be helpful when adding integers with larger absolute values.

- Complete each statement based on your understanding of adding integers so far.

- When adding two positive integers, the sum is

☒ positive.
☐ negative.

- When adding two negative integers, the sum is

☐ positive.
☒ negative.

When adding a positive and a negative integer, the sum will be either positive or negative, depending on which value has the larger absolute value.

- Consider the expression $40 + (-165)$.

- What will be the sign of the sum? ☐ positive
The sign of the sum will be negative. ☒ negative
- Explain your reasoning.
The sum will be negative because -165 has the greater absolute value.

When adding these integers, zero pairs are created.

- Use additive inverses to explain how many zero pairs are created.
40 zero pairs are created. There are the 40 positive integers, from the addend 40, and 40 negative integers, from the -165 addend, that can be combined to form zero pairs.
- What remains after combining the additive inverses?
125 negative integers remain.
- Complete the equation.

$$40 + (-165) = -125$$

- Complete the statements to describe how to find the sum of $-1,029 + (-38)$.

When adding $-1,029 + (-38)$, ☐ zero pairs
☒ no zero pairs are created because the signs of the addends are

☒ the same.
☐ opposites.

The sum can be found by adding the absolute values of the addends. Since both addends are ☐ positive,
☒ negative, the ray for the second addend of the sum will go farther to the ☐ right ($\rightarrow$)
☒ left ($\leftarrow$) of zero on a number

line.

The sum of $-1,029 + (-38)$ is

☐ 991.
☐ -991.
☐ 1067.
☒ -1067.

**7.5.4 Try It: Adding Integers**

1. Find each sum.

a. $-119 + 99$

 -20

c. $-853 + (-505)$

 -1358

b. $254 + (-610)$

 -356

d. $-117 + 110$

 -7 **7.5.4 TRY IT: ADDING INTEGERS**

Component Overview: Students practice adding integers with the same and different signs. Consider using the Think-Pair-Share routine to provide an auditory entry point. The partner discussions will help students gain confidence and understanding.

QUESTIONING

For students who might finish early, consider challenging them by providing more examples of adding integers.

DIFFERENTIATE INSTRUCTION

For students in need of lower entry points, consider allowing them to substitute their own integers to use or give them integers to choose from between 1 and 20.

**7.5.5 Collaboration: Reasoning with Integers**

1. Work with your partner to fill in the blanks with an integer to make each equation true.

$$14 + \boxed{-49} = \boxed{-35}$$

$$\boxed{304} + \boxed{-70} = \boxed{234}$$

$$\boxed{-136} + \boxed{161} = \boxed{25}$$

2. Choose one equation and explain your thinking in determining the missing value.

Student choice will vary.

$$14 + \boxed{-49} = \boxed{-35}$$

Since 14 is positive and the sum is negative, the other addend must be negative. The sign of the sum indicates which addend is the integer with the greater absolute value. Therefore, the missing addend will be negative.

To determine the missing value, I added the absolute values of both existing numbers in the equation, so that zero pairs can be created. Then, I kept the sign of the sum. $|14| + |-35| = 49$. The missing addend is therefore -49 .

7.5.5 COLLABORATION: REASONING WITH INTEGERS

Component Overview: Working with a partner, students determine the missing addends in integer addition equations. Students finish the activity by explaining how they determined the missing addends.

TEACHER MOVES**Turn and Talk**

This routine provides students an opportunity to share their thoughts, reflect, and finalize their answers. To increase active listening, consider asking partners to explain each other's work.

7.5.6 YOUR TURN!

Component Overview: Students develop procedural fluency adding integers. Students work with one-, two-, and three-digit numbers.

ENRICHMENT

Prompt students with the following:

- How do the absolute values of the addends help you determine the sign of the sum?
- How many zero pairs are created when adding these values?
- What remains after eliminating the zero pairs?



7.5.6 Your Turn!

1. Find each sum.

a. $34 + (-21)$

13

b. $-67 + 40$

-27

c. $-333 + (-509)$

-842

d. $700 + (-814)$

-114

e. $-391 + 393$

2

f. $-671 + (-405)$

-1,076



7.5.7 Wrap-Up: Growth Mindset

1. Today, I am still unsure about _____.

2. To fill in this gap, I intend to _____.

Answers will vary.

7.5.7 WRAP-UP: GROWTH MINDSET

Component Overview: Students share what they are still unsure about and develop a plan for how they can fill in the gap(s). Consider using students' perceptions of their understanding, along with their accuracy on the questions in 7.5.6 Your Turn!, to determine which students need support before the next lesson.

DIFFERENTIATE INSTRUCTION

Consider providing answer frames for students to select all that apply.

- The meaning of additive inverse
- The meaning of absolute value
- How to use integer chips to represent zero pairs
- Fluency adding integers with the same sign
- Fluency adding integers with opposite signs